Capstone Project Part 3.3

Part 1: Technical Specifications Impact

Base Prompt

```
portrait of a young woman with blonde hair
```

Technical Prompt

```
portrait of a young woman with blonde hair, shot on Hasselblad H6D,
80mm lens at f/2.8, natural window light, Kodak Portra 400 film
simulation, shallow depth of field
```

Pair 2: City Street

Base Prompt

```
modern city street at night
```

Technical Prompt

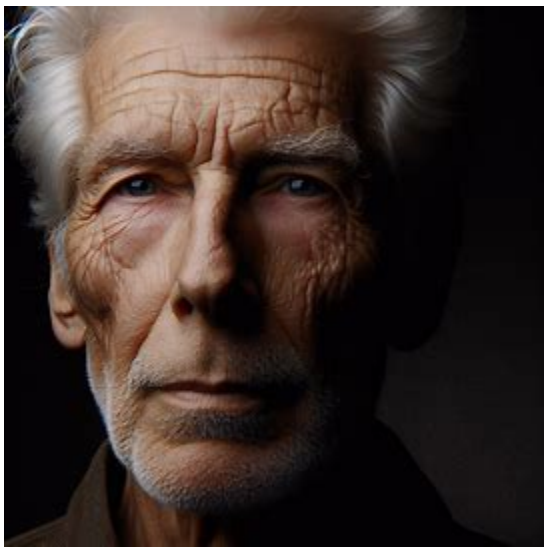
```
modern city street at night, Sony A7R IV, 24mm f/1.4 lens, long
exposure, high contrast, neon reflections on wet pavement, 8K
resolution
```

Part 2: Lighting Experiments

🔹 Prompts Used

1. **Natural Light**
   `elderly man portrait, natural light`

2. **Studio Lighting (Split Lighting)**
   `elderly man portrait, split lighting setup, black background, studio strobe with 36-inch octabox`



3. **Cinematic Golden Hour**
   `elderly man portrait, golden hour sunlight through window, dust`

particles visible, cinematic, Arri Alexa

Part 3: Style and Medium Control (6 minutes)

Photography: mountain landscape, morning mist, shot on Phase One IQ4, 150mm lens, HDR bracketing

Digital Art: mountain landscape, morning mist, digital art, 4K concept art, matte painting style, atmospheric perspective

Traditional Medium: mountain landscape, morning mist, oil painting on canvas, impressionist style, visible brushstrokes

1. Impact of Technical Terms on Medium

- **Photography Prompt**
  - Terms like *Phase One IQ4, 150mm lens, HDR bracketing* produced a **realistic, high-dynamic-range image**
  - Clear depth compression and natural lighting behavior
  - Output closely resembles professional landscape photography
- **Digital Art Prompt**
  - Terms like *concept art, matte painting, atmospheric perspective* resulted in a **stylized, cinematic environment**
  - Enhanced color grading and exaggerated depth
  - Less realism, more artistic interpretation
- **Traditional Medium Prompt**
  - Terms like *oil painting, impressionist, brushstrokes* produced a **painterly texture**
  - Visible strokes and softer edges
  - Reduced sharpness, increased artistic abstraction

2. Which Terms Work Best

- **Photography**
  - Camera models (Phase One, Sony, Hasselblad)
  - Lens specifications (mm, aperture)
  - Lighting techniques (HDR, natural light)
- **Digital Art**
  - Style keywords (concept art, matte painting)
  - Visual techniques (atmospheric perspective, cinematic)
  - Resolution cues (4K, ultra-detailed)
- **Traditional Art**
  - Medium (oil painting, watercolor)
  - Style (impressionist, realism)
  - Texture descriptors (brushstrokes, canvas)

3. Consistency of Results

- **Photography**
  - Most consistent and predictable
  - Strong adherence to realism
- **Digital Art**
  - Moderately consistent

    - Variation in artistic interpretation
- **Traditional Medium**
    - Least consistent
    - Style varies depending on model interpretation of “impressionist”

4. Key Insight

- Technical terms act as **style controllers**
- Camera terms: realism
- Art terms: stylization
- Medium terms: texture and abstraction

Conclusions

Best Practices Discovered

The experiments demonstrate that prompt engineering significantly influences the quality, realism, and consistency of generated images. The most effective approach is to use **clear, descriptive, and structured prompts** that guide the model across multiple dimensions such as subject, lighting, composition, and style. Replacing vague or generic terms with **specific attributes** improves control and reduces ambiguity. Additionally, avoiding conflicting instructions and reframing prompts positively instead of relying on negative constraints leads to more coherent outputs.

Another key practice is the use of **domain-specific language**, such as photography and artistic terminology. These terms help the model align outputs with real-world visual principles, resulting in more professional and predictable results.

Optimal Prompt Structures

An effective prompt follows a structured format that progressively refines the image generation process:

Subject > Environment > Lighting > Technical Details > Style > Quality

For example:

- Subject: portrait of a young woman
- Environment: indoor, near window
- Lighting: natural soft light
- Technical details: 85mm lens, f/2.8
- Style: cinematic photography
- Quality: high dynamic range, 8K

This structured approach reduces randomness and ensures that all critical aspects of the image are explicitly controlled. Prompts that follow this sequence consistently produce more refined and visually coherent outputs compared to unstructured descriptions.

Extension Challenge

```
still life of vintage cameras arranged on a worn wooden table, Leica
M3, Rolleiflex, and Canon AE-1, soft directional window light from the
left, subtle fill light from reflector, shot on Phase One XF IQ4, 80mm
lens at f/4, shallow depth of field, Kodak Portra 400 film aesthetic,
rich tonal gradation, realistic metal and leather textures, gentle
dust particles in the air, cinematic shadows, high dynamic range, fine
detail, soft color contrast, professional product photography,
editorial still life, subtle film grain, carefully controlled
highlights, premium post-processing, 8K
```

Why These Terms Were Chosen

Lighting setup

- **soft directional window light from the left** gives realistic product-lighting behavior
- **subtle fill light from reflector** prevents overly harsh shadows
- **cinematic shadows** adds mood without losing object detail

Camera/lens choice

- **Phase One XF IQ4** suggests premium medium-format clarity

- **80mm lens at f/4** is suitable for still life, with realistic compression and selective focus

Film or digital aesthetic

- **Kodak Portra 400 film aesthetic** adds warmth, soft contrast, and a refined nostalgic tone
- **subtle film grain** supports the vintage subject matter

Post-processing hints

- **carefully controlled highlights**
- **rich tonal gradation**
- **soft color contrast**
- **premium post-processing**
- **high dynamic range**

These terms help the image feel polished and professional rather than flat or overly synthetic.

The strongest prompt is not just descriptive. It is layered in a logical order:

Subject > Arrangement > Lighting > Camera Specs > Aesthetic > Texture > Post-processing > Quality